**9. Scenario: You work for a real estate agency and have been given a dataset containing information about properties for sale. The dataset is stored in a Pandas DataFrame named property_data. The DataFrame has columns for property ID, location, number of bedrooms, area in square feet, and listing price. Your task is to analyze the data and answer specific questions about the properties. Question: Using Pandas DataFrame operations, how would you find the following information from the property_data DataFrame:**

**1.The average listing price of properties in each location.**
**2.The number of properties with more than four bedrooms.**
**3.The property with the largest area.**

**AIM:**

To analyze property data and find the average listing price in each location, the number of properties with more than four bedrooms, and the property with the largest area using Pandas.

**PROCEDURE:**

1. Import the Pandas library.
2. Create a DataFrame containing property details.
3. Group properties by location and calculate the average listing price.
4. Count properties with more than four bedrooms.
5. Find the property with the largest area using idxmax().
6. Display the results.

**PROGRAM CODE:**

```
import pandas as pd
property_data = pd.DataFrame({
    'Property_ID': ['P1', 'P2', 'P3', 'P4', 'P5'],
    'Location': ['Chennai', 'Mumbai', 'Chennai',
                 'Delhi', 'Mumbai'],
    'Bedrooms': [3, 5, 4, 6, 2],
    'Area': [1200, 2500, 1800, 3000, 1000],
    'Price': [5000000, 9000000, 6500000,
              12000000, 4000000]
})
print("Average Listing Price by Location:")
print(property_data.groupby('Location')['Price'].mean())
print("\nProperties with More Than 4 Bedrooms:")
print((property_data['Bedrooms'] > 4).sum()
largest = property_data.loc[property_data['Area'].idxmax()]
print("\nProperty with Largest Area:")
print(largest)
```

**INPUT:**

| Property_ID | Location | Bedrooms | Area (sq. ft.) | Price |
|---|---|---|---|---|
| P1 | Chennai | 3 | 1200 | 5000000 |
| P2 | Mumbai | 5 | 2500 | 9000000 |
| P3 | Chennai | 4 | 1800 | 6500000 |
| P4 | Delhi | 6 | 3000 | 12000000 |
| P5 | Mumbai | 2 | 1000 | 4000000 |

**OUTPUT:**

```
print("\nProperty with Largest Area:")
print(largest)
```

```
Average Listing Price by Location:
Location
Chennai     5750000.0
Delhi      12000000.0
Mumbai      6500000.0
Name: Price, dtype: float64

Properties with More Than 4 Bedrooms:
2

Property with Largest Area:
Property_ID          P4
Location          Delhi
Bedrooms              6
Area               3000
Price          12000000
Name: 3, dtype: object
```

**RESULT:**

Thus, the average listing prices by location, the count of properties with more than four bedrooms, and the largest property by area were successfully identified.